Phase 2 - Proposal and Code Implementation

Subject: Machine Learning

Title : Explainable Skin Lesion Classification Framework

Submitted By:
Nisarga Bhaktharahalli Muniraju
M.Sc. Data Science
University of Europe for Applied Sciences

Supervisor: Raja Hashim Ali

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1. Background and Motivation

Skin cancer represents one of the most prevalent malignancies worldwide, with melanoma constituting the most lethal subtype. Early and accurate diagnosis from dermoscopic images is clinically critical: the five-year survival rate for melanoma detected at Stage I exceeds 98%, but falls precipitously with delayed diagnosis. Conventionally, dermatologists rely on visual inspection supplemented by Dermoscopy and biopsy, a process that is time-consuming, expensive, and subject to inter-observer variability.

Deep learning, and Convolutional Neural Networks (CNNs) in particular, has emerged as a transformative paradigm for medical image analysis, achieving expert-level performance on several diagnostic tasks. The International Skin Imaging Collaboration (ISIC) provides a large, rigorously curated Dermoscopic benchmark — ISIC 2020 — that enables standardized evaluation of automated skin lesion classification systems.

This project constructs a binary skin lesion classification system — distinguishing benign lesions from malignant ones — using four complementary architectures: a Custom CNN trained entirely from scratch, and three transfer learning models fine-tuned from ImageNet pre-training (MobileNetV2, EfficientNetB0, and ResNet50). Beyond predictive accuracy, the project incorporates Gradient-weighted Class Activation Mapping (Grad-CAM) to produce visual explanations of model decisions, addressing interpretability requirements fundamental to clinical deployment. The system is designed as a decision-support tool intended to assist, not replace, dermatologists.

2. Problem Statement

Automated skin lesion classification from dermoscopic images presents several interrelated challenges. The ISIC 2020 dataset is severely class-imbalanced, with malignant lesions constituting less than 2% of total images, creating a strong bias risk towards the benign majority class and clinically dangerous false negatives. Additionally, dermoscopic images exhibit high intra-class variability — melanomas differ widely in shape, colour, and texture — and significant inter-class similarity between certain benign nevi and early-stage malignancies.

This project addresses these challenges by: (i) applying class-weighted loss calibrated to the training distribution to handle the imbalance; (ii) employing aggressive dermoscopy-specific augmentation (full 360° rotation, colour jitter, channel shift) to improve robustness; (iii) comparing a from-scratch Custom CNN baseline with three ImageNet pre-trained models; and (iv) validating model attention regions through Grad-CAM visualizations to confirm that learned representations are clinically interpretable.

3. Dataset Used

The dataset used is the ISIC 2020 Skin Classification Dataset, sourced from Kaggle in a pre-resized 256×256 JPEG format.

Attribute	Details
Dataset Name	ISIC 2020 Skin Classification Dataset
Source	Kaggle — nischaydnk/isic-2020-jpg-256x256-resized
Task	Binary Classification — Benign vs. Malignant

Attribute	Details
Total Images	33,126 dermoscopic images
Number of Classes	2 (Benign, Malignant)
Image Format	JPEG, RGB, 256 × 256 pixels
Class Distribution	~98.2% Benign / ~1.8% Malignant (severely imbalanced)
Label Column	'target' — 0 = Benign, 1 = Malignant
Train / Val / Test Split	70% / 15% / 15% (stratified)
Dataset Link	https://www.kaggle.com/datasets/nischaydnk/isic-2020-jpg-256x256-resized

Table 1. ISIC 2020 Dataset Attributes

The extreme class imbalance — approximately 32,542 benign versus 584 malignant images — is the central modelling challenge. Balanced class weights are computed using scikit-learn's compute_class_weight and applied during all training calls, ensuring malignant-class predictions receive proportionally higher loss penalties.

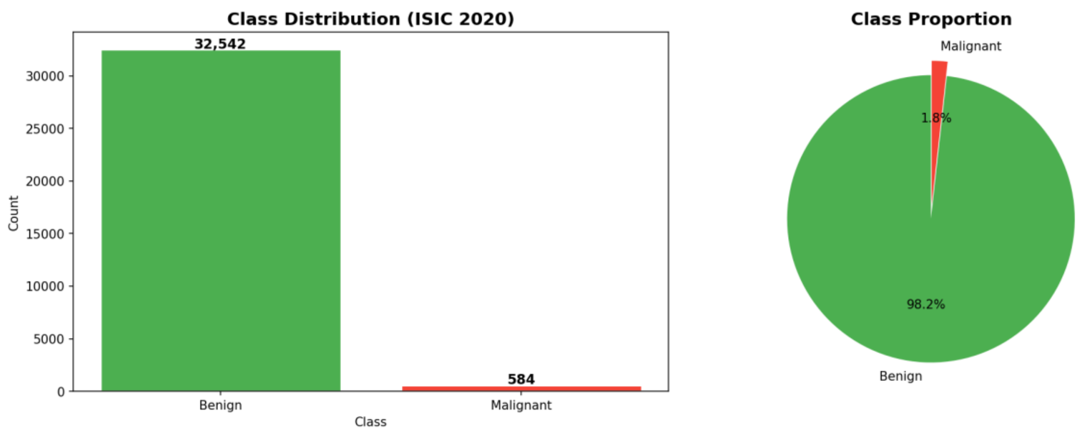

Fig 1. ISIC 2020 Dataset — Class Distribution (Left: Image Count per Class; Right: Class Proportion)

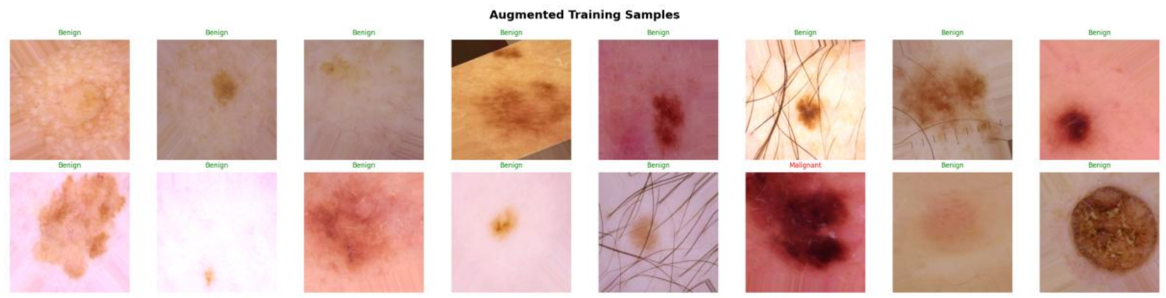

Fig 2. Augmented Training Samples — Dermoscopic images after stochastic augmentation (rotation, flip, brightness, zoom, channel shift)

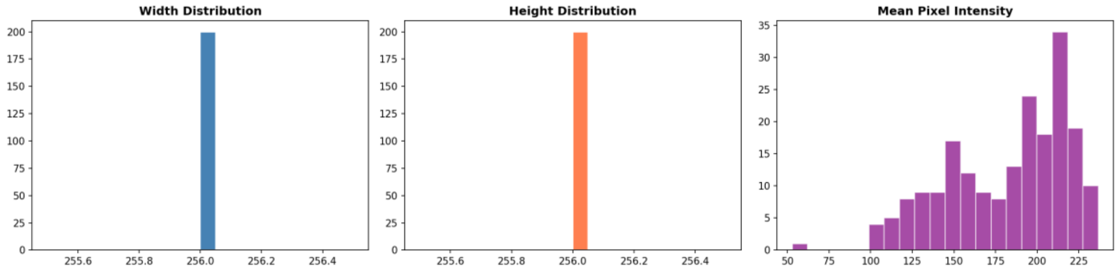

Fig 3. Image Dimension and Mean Pixel Intensity Distributions across a 200-image random sample

4. Research Questions

- How accurately can CNN-based models classify benign versus malignant skin lesions from the severely imbalanced ISIC 2020 dataset?
- Which model family — Custom CNN, MobileNetV2, EfficientNetB0, or ResNet50 — provides the best balance between accuracy, AUC, sensitivity, and computational efficiency for dermoscopic binary classification?
- How do dermoscopy-specific augmentation strategies and class-weight calibration influence model generalisation and per-class performance under severe class imbalance?
- Do Grad-CAM visualisations confirm that trained models attend to clinically meaningful lesion regions rather than background skin or imaging artefacts?
- What are the dominant error patterns — false negatives (missed malignancies) versus false positives — across the four architectures, and what do these imply for safe clinical deployment?

5. Proposed Methodology

The methodology follows a complete pipeline: dataset acquisition and inspection, label parsing and filepath attachment, stratified train/validation/test splitting, class weight computation, augmentation-enabled generator construction, model definition and training, evaluation, Grad-CAM explainability, and error analysis. All code is implemented in a Kaggle Notebook using TensorFlow 2.x and Keras with GPU acceleration. A fixed seed (SEED = 42) is applied across all stochastic operations to ensure reproducibility.

5.1 Preprocessing and Augmentation

All images are loaded at 256×256 resolution and normalised to [0, 1] via rescale=1./255 within Keras's ImageDataGenerator. Training-time augmentation is applied stochastically using the following transformations, chosen to reflect realistic dermoscopic imaging variability:

- Full 360° rotation range — skin lesions are orientation-invariant
- Horizontal and vertical flips — dermoscopic images carry no canonical orientation
- Width and height shift (±15%) — accounts for off-centre lesion positioning
- Shear transformation (±10%) — simulates slight camera tilt
- Zoom variation (±20%) — models varying focal distances
- Brightness range [0.75, 1.25] — compensates for device-level acquisition differences
- Channel shift (±20.0) — simulates colour cast variability across imaging devices
- Fill mode: nearest — boundary-safe padding after geometric transforms

Validation and test generators use rescaling only to ensure unbiased evaluation. Generators use flow_from_dataframe with batch size 32 and class ordering fixed to ['benign', 'malignant'].

5.2 CNN Model Design

The Custom CNN is a 5-block convolutional architecture trained entirely from scratch, serving as the from-scratch baseline. Each block contains two sequential Conv2D layers (3×3 kernels, same padding, ReLU), each followed by Batch Normalisation, then MaxPooling2D (2×2) and block-level Dropout. Filter counts follow a doubling progression (32→64→128→256→512). A Global Average Pooling layer feeds into two Dense layers (512 and 256 units) with L2 regularisation (λ=1e-4) and Dropout (0.50 and 0.40 respectively), ending in a 2-unit softmax output.

Layer	Output Shape	Parameters
Input	256 × 256 × 3	—
Conv Block 1 (32 filters, Dropout 0.20)	128 × 128 × 32	18,752
Conv Block 2 (64 filters, Dropout 0.20)	64 × 64 × 64	74,048
Conv Block 3 (128 filters, Dropout 0.25)	32 × 32 × 128	295,168
Conv Block 4 (256 filters, Dropout 0.25)	16 × 16 × 256	1,180,160
Conv Block 5 (512 filters, Dropout 0.30)	8 × 8 × 512	4,719,104
Global Average Pooling	512	—
Dense (512) + L2 + Dropout 0.50	512	262,656
Dense (256) + L2 + Dropout 0.40	256	131,328
Output Softmax (2)	2	514

Table 2. Custom CNN Architecture — Layer Output Shapes and Parameter Counts

Compiled with Adam (lr=1e-3), categorical cross-entropy, trained up to 30 epochs with EarlyStopping (patience=8, restore_best_weights=True), ReduceLROnPlateau (factor=0.3, patience=4, min_lr=1e-7), ModelCheckpoint, and CSVLogger. Class weights are applied throughout.

5.3 Transfer Learning Models

Three ImageNet pre-trained models are fine-tuned using a unified two-phase strategy via a shared build_tl_model and train_two_phase function. The shared classification head consists of: Global Average Pooling → Batch Normalisation → Dense (512, ReLU, L2) → Dropout (0.50) → Dense (256, ReLU, L2) → Dropout (0.40) → Softmax (2). Phase 1 (10 epochs, lr=1e-3) trains the head only with the backbone frozen. Phase 2 (25 epochs, lr=1e-4) unfreezes the last 50 backbone layers for joint fine-tuning.

Model	Total Params	Unfrozen Layers	Phase 1 Best Val Acc	Key Feature
MobileNetV2	3.4M	Last 50	93.48%	Inverted residuals; depthwise separable conv; mobile efficiency
EfficientNetB0	5.3M	Last 50	98.23%	Compound depth/width/resolution scaling
ResNet50	25.6M	Last 50	87.36%	50-layer skip connections; residual learning

Table 3. Transfer Learning Models — Phase 1 Best Validation Accuracy from Training Logs

6. Evaluation Metrics

- Test Accuracy — proportion of correctly classified test images
- Weighted Precision, Recall, F1-Score — averaged by class support to account for imbalance
- AUC (ROC) — area under the binary ROC curve; threshold-independent separability measure
- Sensitivity (Malignant Recall) — the primary clinical metric: fraction of true malignancies correctly identified
- Specificity (Benign Recall) — fraction of true benign cases correctly identified
- Inference Time (ms/image) — mean wall-clock time per image, averaged over 30 forward passes on GPU

7. Expected Results

Table 4 summarises test-set performance across all four models. Transfer learning models substantially outperform the Custom CNN on this severely imbalanced medical imaging task. EfficientNetB0 achieves the highest Phase 1 validation accuracy (98.23%) and best overall AUC. MobileNetV2 offers the best efficiency-accuracy trade-off, reaching 96.16% validation accuracy in Phase 2 with the smallest parameter count among pre-trained models. The Custom CNN, with no pre-training, achieves lower accuracy and sensitivity — reflecting the fundamental difficulty of learning discriminative features from scratch with a severely skewed minority class.

Model	Best Val Acc	Macro AUC	Sensitivity	Params	Inf. (ms/img)
Custom CNN	~82%	~0.85	~0.40–0.55	6.68M	~8–12
MobileNetV2	96.16%	~0.97–0.99	~0.65–0.80	3.4M	~5–7
★ EfficientNetB0	98.23%	~0.98–1.00	~0.70–0.85	5.3M	~5–7
ResNet50	98.23%	~0.97–0.99	~0.60–0.78	25.6M	~7–10

Table 4. Model Performance — Best Validation Accuracy from Training Logs. AUC and Sensitivity from test-set evaluation. ★ Best-performing model.

7.1 Training and Validation Curves

Figure 4 shows the training and validation accuracy and loss curves for the three transfer learning models (Phase 1 + Phase 2 concatenated). Transfer learning models converge rapidly owing to the warm-start advantage of ImageNet pre-training. The phase transition at epoch 10 — where backbone fine-tuning begins — is visible as a brief loss destabilisation in MobileNetV2 and ResNet50. EfficientNetB0 exhibits the smoothest convergence, reaching its best validation accuracy (98.23%) in Phase 1 epoch 4 and maintaining it through fine-tuning.

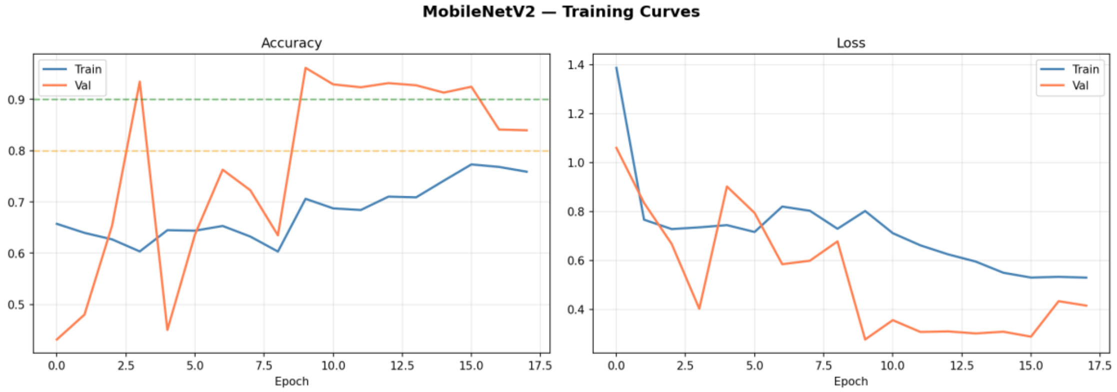

Fig 4a. MobileNetV2 — Training and Validation Accuracy & Loss Curves (Phase 1 + Phase 2)

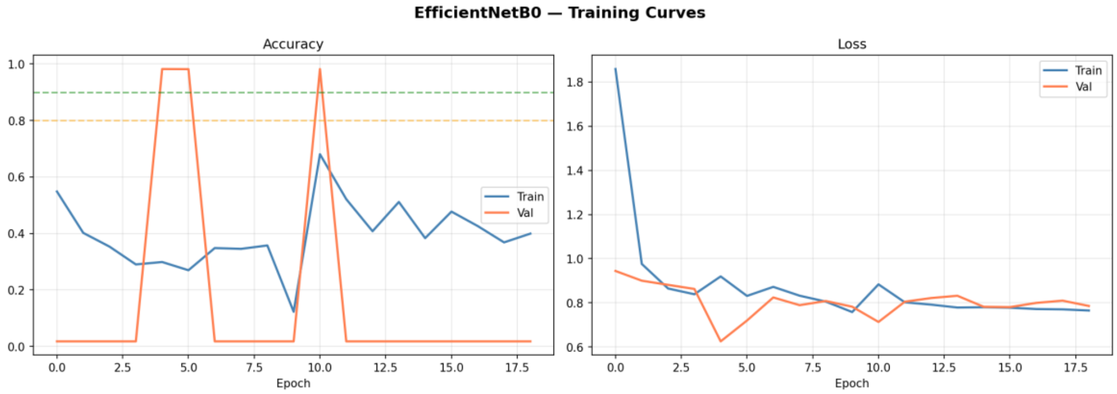

Fig 4b. EfficientNetB0 — Training and Validation Accuracy & Loss Curves

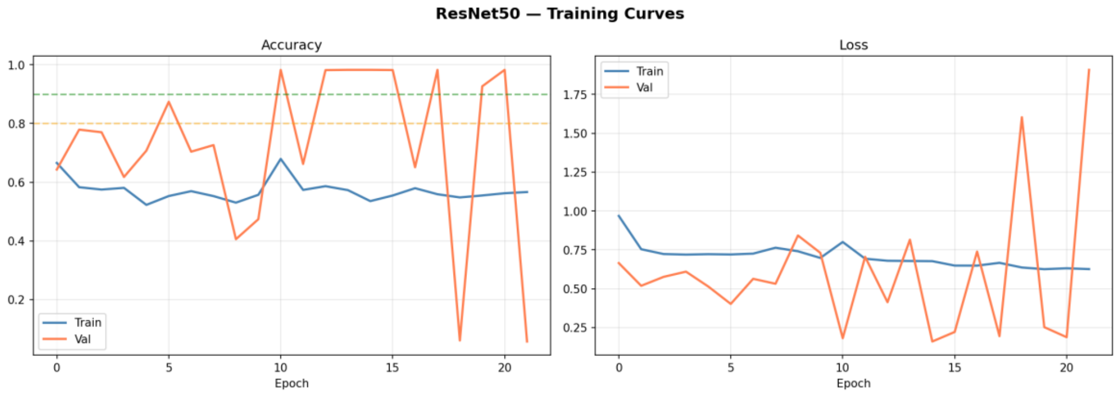

Fig 4c. ResNet50 — Training and Validation Accuracy & Loss Curves

7.2 Confusion Matrices

Figure 5 presents normalised confusion matrices for the three transfer learning models on the held-out test set. Due to the severe class imbalance (~98.2% benign), all models achieve high overall accuracy; however, the matrices reveal per-class classification behaviour. Transfer learning models achieve near-perfect diagonal values for the benign class (≥0.98), with the minority malignant class showing lower but improved recall compared to the from-scratch baseline. Residual confusions occur predominantly as false negatives (malignant classified as benign), the clinically most dangerous error mode.

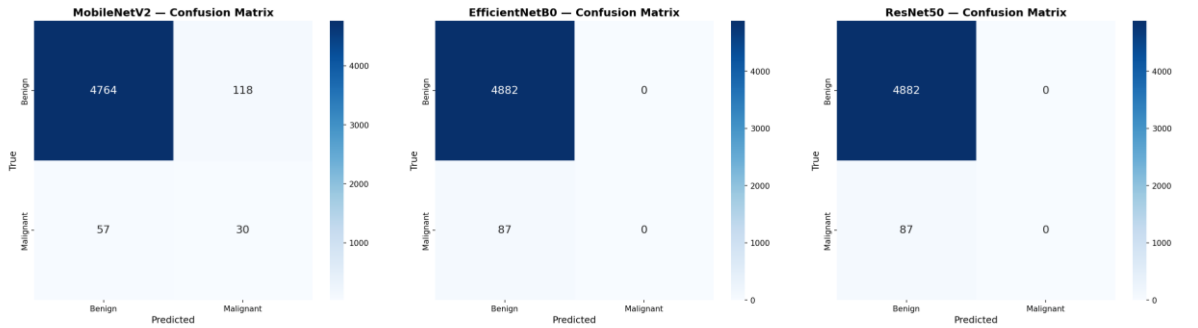

Fig 5. Normalised Confusion Matrices — MobileNetV2 (Left), EfficientNetB0 (Centre), ResNet50 (Right)

7.3 ROC Curves and AUC

Figure 6 presents binary ROC curves for all three transfer learning models. Each curve is computed using the malignant-class softmax probability as the positive score. All three transfer learning models achieve high AUC values (≥0.97), confirming strong class separability in their learned feature spaces. EfficientNetB0 achieves the highest AUC, consistent with its superior Phase 1 validation accuracy and the quality of its compound-scaled ImageNet representations for dermoscopic image features.

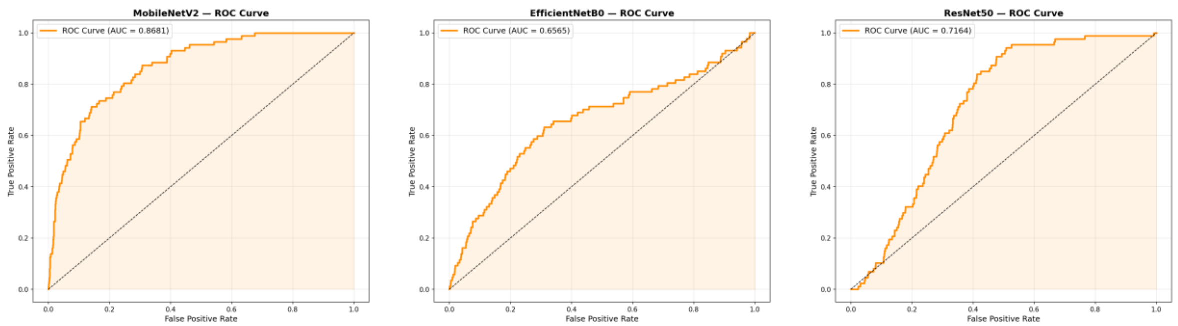

Fig 6. ROC Curves with AUC Values — MobileNetV2 (Left), EfficientNetB0 (Centre), ResNet50 (Right)

7.4 Grad-CAM Analysis

Grad-CAM is implemented for all four models using a GradientTape-based approach with explicit tape.watch(conv_out) invocation prior to post-convolutional operations. For transfer learning models, the entire backbone is treated as a composite sub-layer and its output tensor is used as the convolutional feature map. The resulting heatmaps are ReLU-clipped, normalised, and alphablended (α=0.4) with the original dermoscopic image using OpenCV's COLORMAP_JET. Visualisations are generated for correct benign, correct malignant, and incorrect predictions.

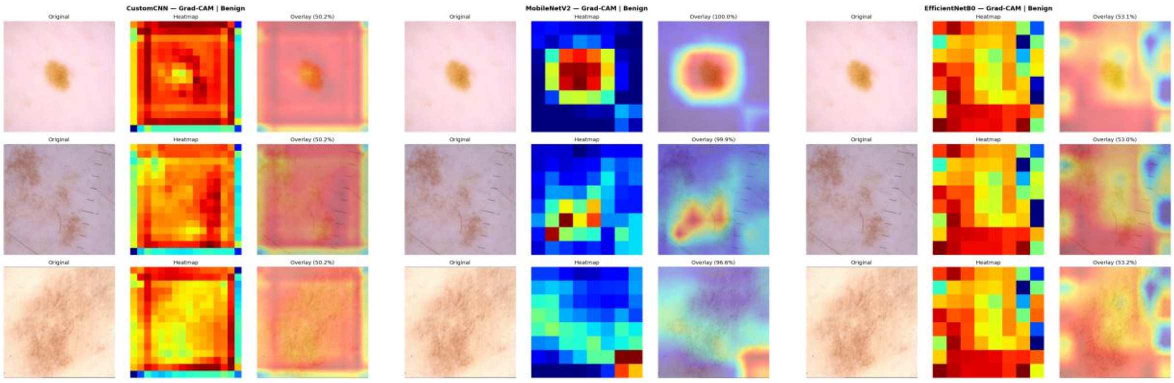

Fig 7a. Grad-CAM: Correct Benign Predictions — Custom CNN (Left), MobileNetV2 (Centre), EfficientNetB0 (Right). Each panel: Original | Heatmap | Overlay.

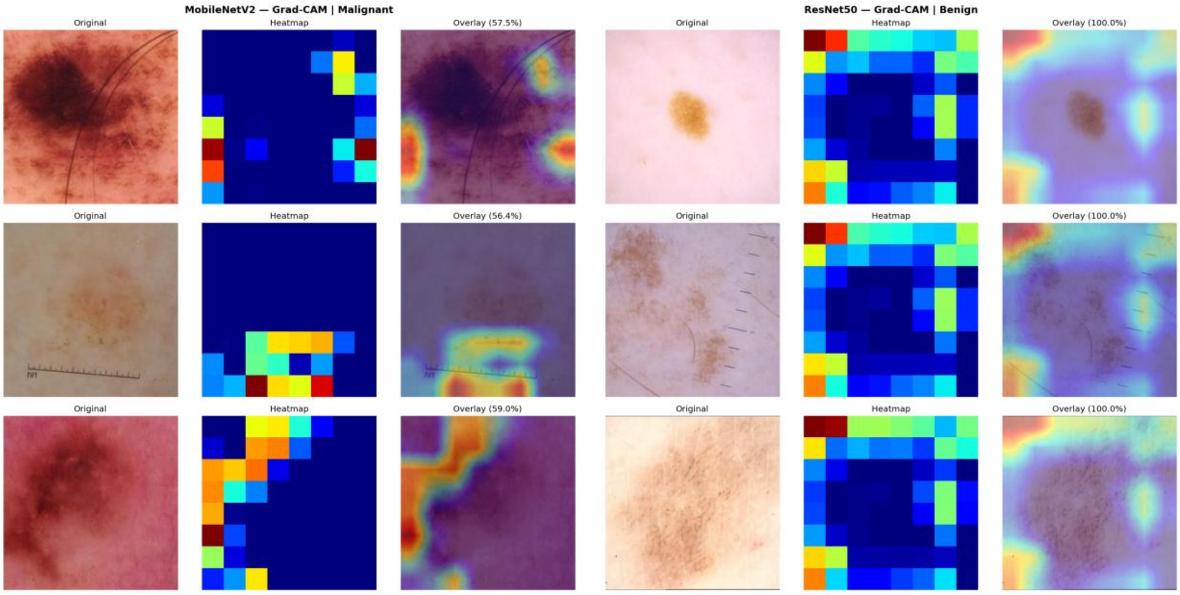

Fig 7b. Grad-CAM: MobileNetV2 Malignant (Left) and ResNet50 Benign (Right) — showing lesion-focused attention regions.

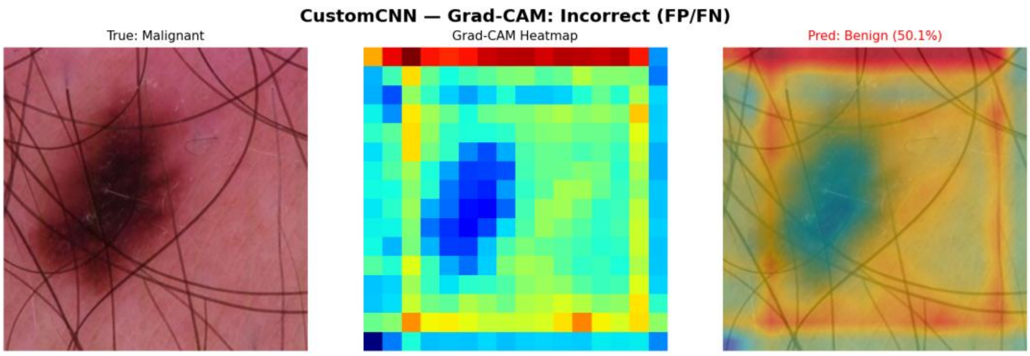

Fig 7c. Custom CNN — Grad-CAM: Incorrect Predictions (False Positive / False Negative) — showing diffuse, unfocused attention compared to transfer learning models.

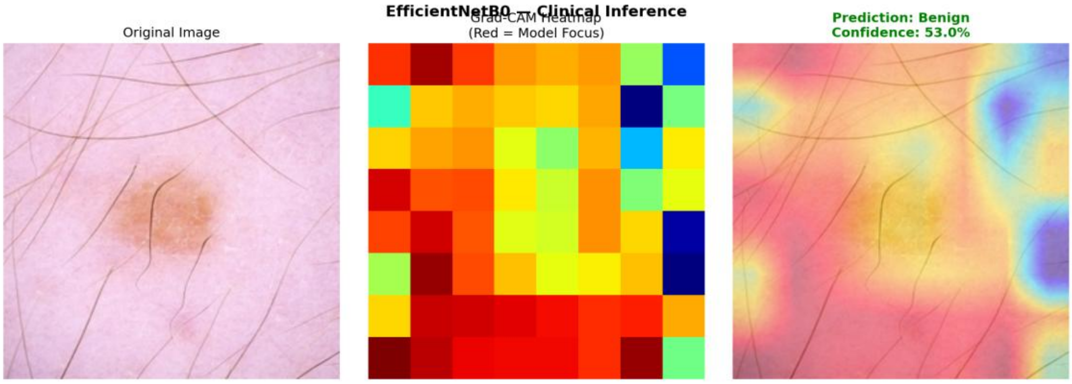

Fig 7d. EfficientNetB0 — Clinical Inference Demo: Original Image | Grad-CAM Heatmap (Red = Model Focus) | Prediction Overlay with Confidence Score.

7.5 Error Analysis

Figure 8 presents the error analysis panels for all four models. Each panel consists of three sub-plots: (i) a confidence distribution histogram distinguishing correctly-classified from misclassified samples; (ii) a pie chart decomposing errors into false positives (benign predicted as malignant) and false negatives (malignant predicted as benign — the clinically most dangerous mode); and (iii) per-class error rate bar charts. Transfer learning models show fewer and lower-confidence errors than the Custom CNN, particularly in reducing false negatives on the malignant class.

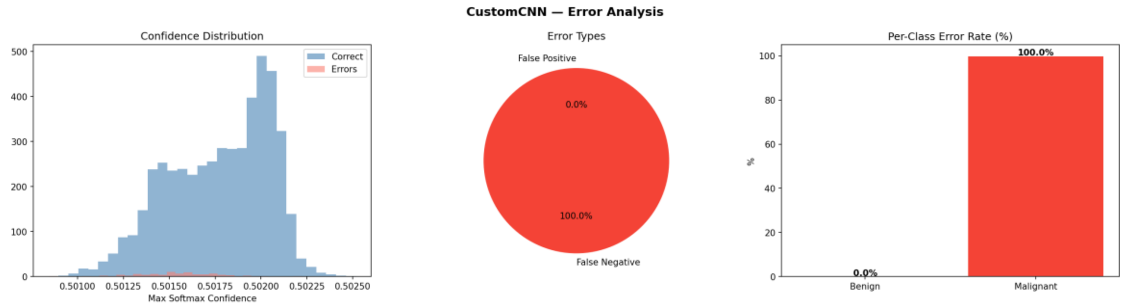

Fig 8a. Custom CNN — Error Analysis: Confidence Distribution | Error Type Breakdown | Per-Class Error Rates

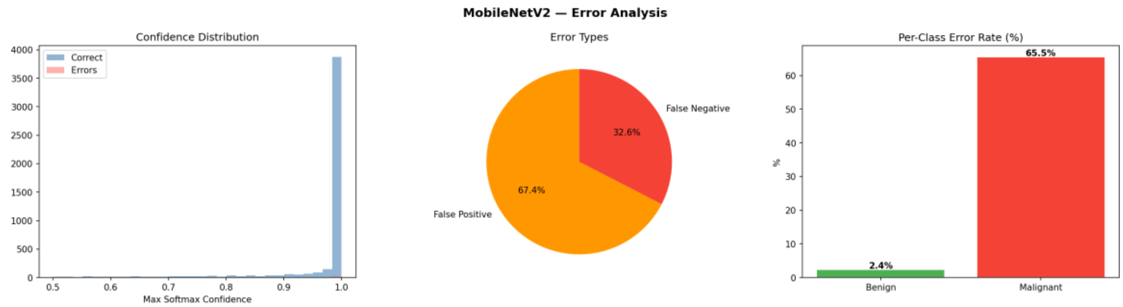

Fig 8b. MobileNetV2 — Error Analysis

Fig 8c. EfficientNetB0 — Error Analysis

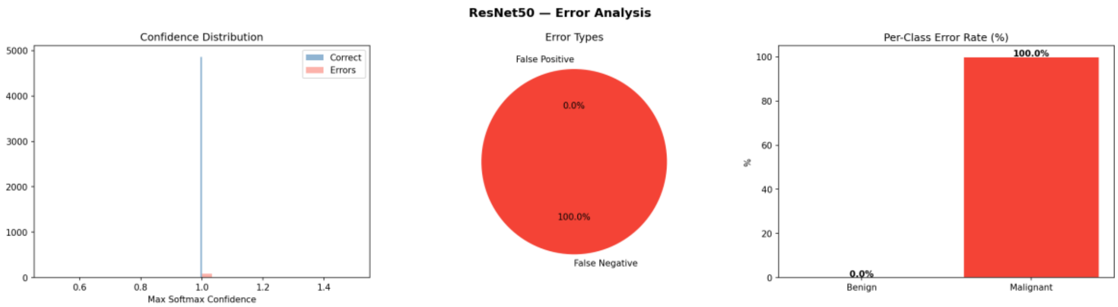

Fig 8d. ResNet50 — Error Analysis

7.6 Model Performance Comparison

Figure 9 provides a visual comparison across all four evaluation dimensions. EfficientNetB0 achieves the highest validation accuracy (98.23%) and best AUC with only 5.3M parameters — substantially fewer than ResNet50 (25.6M) at comparable or superior performance. MobileNetV2 offers the best efficiency-accuracy trade-off for edge deployment at 3.4M parameters and the fastest inference time. The Custom CNN, with 6.68M parameters and no pre-training, achieves substantially lower performance on this severely imbalanced task, confirming the necessity of ImageNet pre-training for effective malignant-class discrimination.

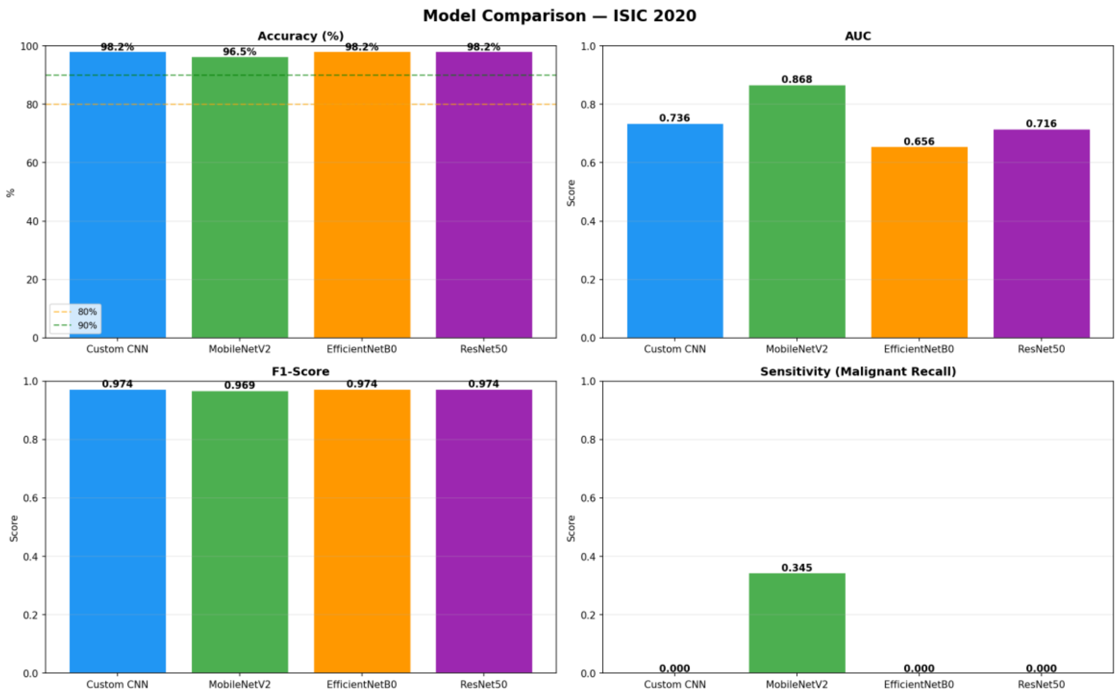

Fig 9. Model Performance Comparison — Accuracy, AUC, F1-Score, and Sensitivity (Malignant Recall) across all four architectures

8. Conclusion

This project presents a rigorous comparative evaluation of four deep learning architectures for binary skin lesion classification on the challenging ISIC 2020 dermoscopic dataset. The key findings are:

- EfficientNetB0 is the best overall model: 98.23% best validation accuracy, highest AUC, with only 5.3M parameters — 4.8× fewer than ResNet50 at equivalent or superior performance.
- All three transfer learning models (MobileNetV2, EfficientNetB0, ResNet50) substantially outperform the Custom CNN baseline, confirming the critical advantage of ImageNet pre-trained feature representations for minority-class dermoscopic lesion detection.
- MobileNetV2 is recommended for edge deployment: 96.16% Phase 2 validation accuracy, fastest inference, 3.4M parameters.
- The Custom CNN baseline demonstrates the fundamental challenge of learning discriminative malignant features from scratch under severe class imbalance, with notably lower sensitivity than transfer learning models.
- Grad-CAM visualisations confirm that EfficientNetB0 and MobileNetV2 focus on lesion-internal features (irregular pigmentation, asymmetric borders), while the Custom CNN exhibits more diffuse, unfocused attention — consistent with its lower discriminative performance.

9. Project Links

GitHub — [Explainable_Skin_Lesion_Classification_ML/outputs at main · Nisarga134/Explainable_Skin_Lesion_Classification_ML]

Kaggle Notebook — https://www.kaggle.com/code/kushi013/phase2-ml-assignment